Deep Learning models in Identification of different classes in Bacteria

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Abstract— In the Medical Field the proper identification of a bacteria can lead to life saving and clinical decision making when it comes to both patient life and medical development. Traditionally this was done via Bacterial identification that required professional medical equipment, operations, and trained employees as one would assume this process can be quite expensive as well as time-consuming. With the development of Deep Learning models, I was curious if the creation of a Convolutional neural network could help with the identification of some of the major types of Bacteria. Multiple models were developed to see if there was a certain approach to hyperparameters that would lead to a road able to best predict the type of bacteria. I created 8 classes of Data and used 4 different models to try and find the best approach to identification of the model.

Keywords—Bacteria, Classification, model, identification, Deep Learning, Convolutional Neural Network.

I. Introduction

In the realm of Microbiology, the identification of the specific types of bacteria that show up in an experiment are almost the most important next to the treatment or development of the product or patient. Historically microbiologist or bacteriologist would do this using a large number of techniques such as morphological, staining characteristics, and mapping of the DNA, all of these techniques would need professionals operation the processing and each possible methodology takes time and can be quite costly for example each Bacteria sent to be identified at a specialty lab has a starting cost of around $300 dollars and hundred will pass through a lab in a day. The time and money needed for this crucial step in the medical field has a massive impact on patient care and medical development, the time waiting for identification has an impact on the quality of care that one might receive or the ability to ensure the safety of a product and its sterility.

Learning based techniques had already been seen entering the medical field to tackle a multitude of issues, the most well-known one being the attempt to train models to identify cancer cells before they become an issue for patients. The initial steps in machine learning required manual extraction of features for parameters and filters and the results were quite weak and limited [1-2]. With the development of CNN, I hope that it will be able to find a ground where it can possibly identify key types of bacteria or at least some of the general classes [3] of bacteria because intra-class differences are nearly impossible to tell just from a normal photo. Multiple studies have shown that spectral information can be important to the identification of a bacteria [4-5]. By using a bi-directional convolutional neural network I believe that that adjustment of weights will be able to give insight into the proper identification of classes of Bacterium based off a simple images to see if it can be turned into one of 8 classes of pre-determined bacterium based up the frequency they are run into in a lab environment and there morphology under a microscope. This will help create a fast-acting and cost-effective alternative when compared to other methodologies spoken about. This method will be capable of identifying spatial and color features of a bacteria to classify it or at least identify the type of bacteria for better judgment of next steps in patient care.

II. Literature Review

A. Background.

The ability to tell apart different types of bacteria has been a crucial part of the medical field since its inception. This can be broken down into a bacteria's classification, characteristics and any other variable that a particular lab may be interested in for their needs and purposes. The initial dive into the medical space for deep learning image models has been quite difficult to get acceptable results either because of how strict the medical field is and the accuracy of the models falling too short for comfort when making such important decisions [6]. May different types of deep learning are currently being tried, from CNN's, reinforcement learning or the use of transformers to optimize already existing techniques. There is currently other paths that are being taken into medical aid such as imaging for radiologist [7] or using deep learning to optimize nucleic acid and peptide processing.

Fig. 1	Type of Bacteria		
	Class	*Quantity*	*% Total*
1	Amoeba	75	12.5%
2	Euglena	75	12.5%
3	Hydra	75	12.5%
4	Paramecium	75	12.5%
5	Spherical	75	12.5%
6	Spiral	75	12.5%
7	Rord	75	12.5%
8	Yeast	75	12.5%

When looking at current development in this field there is not much on taking the image of a bacteria and attempting to identify its class but rather from a later step in the identification process, particularly at the step of DNA sequencing. Due to the large amount of DNA data already available and how it breaks down into the specific types of classes based on historical relationships and DNA it is an optimal source of data to approach this subject by cutting time and cost at the tail end of the process rather than the entire thing.

B. *Other approaches being undertaken*

Within the field of identification via imaging there is a few other methods that are being approached that unfortunately I cannot be due to the method of the gather of the Data. Once such way is looking at the images under different spectrums to see key details about the bacteria that would be missed in normal imagination. There are also techniques of pre-straining the bacteria to pre-deduce the class this is possible because gram straining separates the bacteria into two categories before even attempting identification gram negative and gram positive [8].

III. Data Aquisition

Multiple bacteria growths were found on pre-existing samples; the bacteria were then isolated and growth promoted independently till it was a large enough colony. Once colonies were of sufficient size, a portion of the colony was taken and stained. The bacteria sample was then ID and marked for imaging purposes. Once enough samples of each type were found and categorized. I began to stain them with a normal color stain for imaging purposes. In the end I ended up with 75 images of each of 8 classes of bacteria. Each image was taken using a AmScope ME520 Compound microscope with the digital images taken from the software included. The 8 classes of bacteria that was created were Amoeba, Euglena, Hydra, Paramecium, Yeast and the remaining images are by classes of bacteria defined by their morphology, so they are categorized as rod, spherical, and spiral.

This process was necessary as there is a lack of good open source annotated bacterial detection sets that were of microscopic layers. It was useful this way because all the bacteria found and used are relatively common in a lab setting for my job. The separation by morphology for the remaining classes is because even full identification for those types of bacteria is difficult to separate and can help a clinical lab know what type of resistance they may have based upon their shape. The initial microscopic photos were 512x512x3 and thus needed to be scaled properly in order to try and get the best results from the model. (Fig 1). The Data was then modified to have an appropriate input and scaling for the image this was done by patching the image and then applying the lambda function to ensure all images were treated with the same cleaning.

IV. Methodology

A. *Structure of model*

The Final concurrent neural Network model can be seen in Fig.3. image initially comes in at an already augmented size with random rotations to the images to a give much more robust and accurate training for the model. I used a basic Encoder and Decoder shape for the initial model and began building from there.

B. *Develepment of the model*

I initially attempted a simple model but with many filters because of all the unique shapes bacteria in a class can have. When doing this the model took approximately 12 hrs. for the initial training and the results were not in an "acceptable" range even though I attempted to optimize the parameters multiple times, the time to train even with an early termination to see model progress did not yield effective results. So, the number of filters were reduced per level, but more layers were added to compensate for the complexity of our data set, the initial results gave us a 29% accuracy which essentially reduced our model from a 1/8th chance to a ¼ model. When adding a few key dropout and max pooling layers and changing some of the activations I was able to find a way to get it to above 50% but had an overfitting issue (Fig.2).

The final Model was finished by changing the batch size to 64 and epochs to 20 which increased our accuracy without overfitting. I also added a few more layers that were dense, dropout and another convolutional layer.

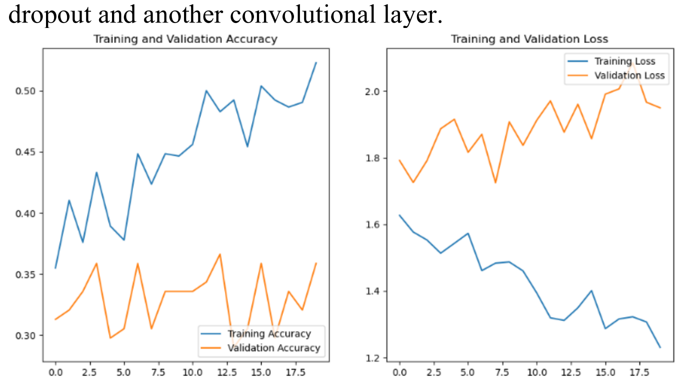

Fig.2 image of Initial Models training and validation accuracy per epoch. Shows over fitting on training when compared to the validation.

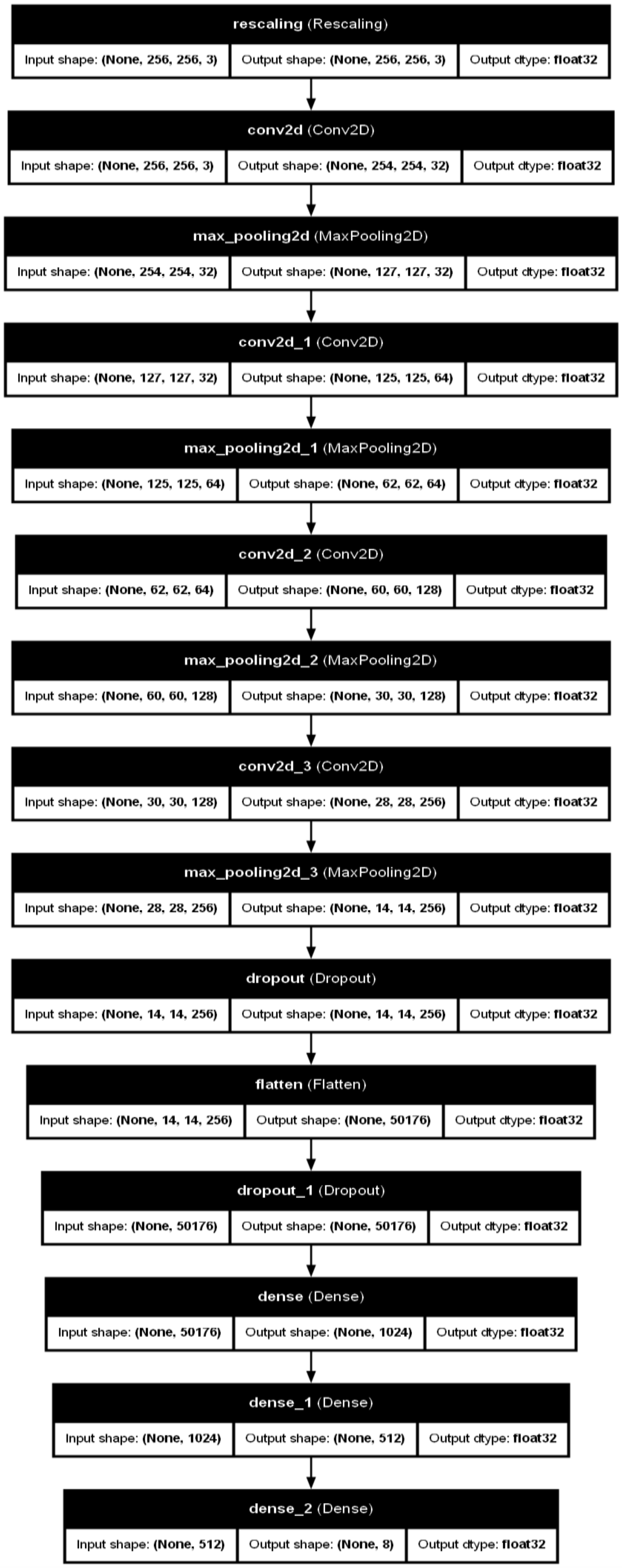

Fig.3 Final Convultional nerual Network with Backpropogation, followds the normal U shape for the most part with overall activations being some form of Relu and finalized with a softmax.

V. Results

The results of the project were able to produce a 32.5% accuracy, increasing the probability from 12.5% to essentially a 1/3 chance(Fig 4.). Although some of the initial models had a higher training accuracy the validation score showed signs of overfitting, so I went with a model I was sure did not have overfitting. Not the best but when compared to other research models that are able to get there models from to the 45-50% range when only implementing R-CNN and not more expensive and time-consuming models [9].

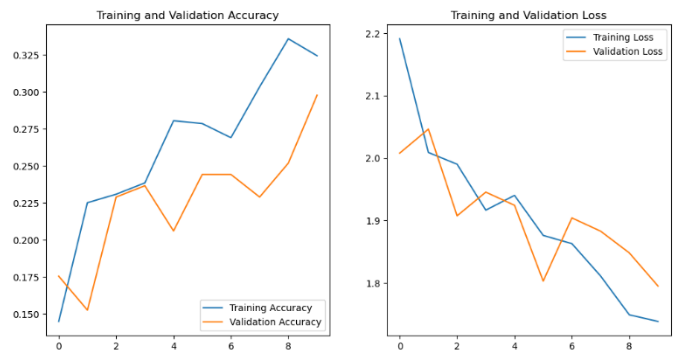

Figure 4. showing training and validation accuracy and loss on final CNN.

VI. Discussions

Looking at other research, I wish I was able to develop and understand the implementation of other means of imaging such as Hyperspectral to show more aspects of the physical characteristics of bacteria to see if that would have such an impact but I was unable to fully grasp an appropriate approach to create it. I also elected to not take a pre-trained model or transfer model when going through this endeavor to see if a much fresher approach would yield better results. I also think I needed more images of each bacterium or a better classification than one normally used based on morphology.

The main point of contention of this research besides the always present time was the lack of resources both physically and to produce, find, and even create the dataset. When gathering the initial data set as stated in the introduction a large amount of time and resources go into the identification of bacteria this ended up taking a lot longer than I initially hoped for because I thought there would be an adequate dataset online that I can use for this model the closest I was able to find was a massive file that was far too complex to try and take on in a reasonable amount of time. Not to mention my computer whether it be how I optimized the system was taking long periods of time per training even leaving me with less than desirable number of permutations that I can go through. The problem that I ran into as well was the fact that I was unable to implement a transformer model to break the images down into patches and from there run it through an embedding process then SoftMax so that it can have a much more global sense of attention for the images to be better defined.

REFERENCES

[1] B. Slabbinck, B. De Baets, P. Dawyndt, et al., “Towards large-scale fame-based bacterial species identification using machine learning techniques,” Syst. Appl. Microbiol. 32, 163–176 (2009).

[2] M. Yang, X. Liu, Y. Luo, et al., “Machine learning-enabled non destructive paper chromogenic array detection of multiplexed viable pathogensonfood,”Nat.Food.2,110–117(2021).

[3] H. Wang, H. Ceylan Koydemir, Y. Qiu, et al., “Early detection and clas sification of live bacteria using time-lapse coherent imaging and deep learning,” Light Sci. Appl. 9, 118 (2020).

[4] Q. Huang, W. Li, B. Zhang, et al., “Blood cell classification based on hyperspectral imaging with modulated Gabor and CNN,” IEEE. J. Biomed.Health.24,160–170(2019).

[5] R. Nißler, O. Bader, M. Dohmen, et al., “Remote near infrared identifi cation of pathogens with multiplexed nanosensors,” Nat. Commun. 11, 5995(2020).

[6] Baron EJ. Classification. In: Baron S, editor. Medical Microbiology. 4th edition. Galveston (TX): University of Texas Medical Branch at Galveston; 1996. Chapter 3. Available from: https://www.ncbi.nlm.nih.gov/books/NBK8406/

[7] Chan HP, Samala RK, Hadjiiski LM, Zhou C. Deep Learning in Medical Image Analysis. Adv Exp Med Biol. 2020;1213:3-21. doi: 10.1007/978-3-030-33128-3_1. PMID: 32030660; PMCID: PMC7442218.

[8] Jennifer Tsang, Ph.D. “Identifying Bacteria through Look, Growth, Stain and Strain.” *ASM.Org*, 7 Feb. 2020, asm.org/articles/2020/february/identifying-bacteria-through-look,-growth,-stain.

[9] Yanik, H., Kaloğlu, A. H., & Değirmenci, E. (2020). Detection of Escherichia coli bacteria in water using deep learning. *Tehnički Glasnik*, *14*(3), 273–280. https://doi.org/10.31803/tg-20200524225359